INCH-POUND

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SUPERSEDING
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8 December 1975

MILITARY SPECIFICATION

SURVEYING INSTRUMENT;

AZIMUTH GYRO, LIGHTWEIGHT

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an automated north seeking gyroscopic survey system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

V-F-106	- Fastener, Slide, Interlocking.
CCC-C-419	- Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army And Numbered.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6675

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MILITARY

MIL-V-173	- Varnish, Moisture And Fungus Resistant (For Treatment of Communications, Electronics And Associated Equipment).
MIL-C-675	- Coating of Glass Optical Elements (Anti-Reflection).
MIL-T-704	- Treatment And Painting Of Materiel.
MIL-S-3534	- Surveying Instruments And Accessories, Packaging Of.
MIL-T-14091	- Tripod, Surveying, Extension-Leg-Type.
MIL-P-19834	- Plate, Identification, Metal Foil, Adhesive Backed.
MIL-C-22751	- Coating System, Epoxy-Polyamide, Chemical And Solvent Resistant: Process For Application Of.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standard For Federal Services.
FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking For Shipment And Storage.
MIL-STD-130	- Identification And Marking Of U.S. Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference.
MIL-STD-810	- Environmental Test Methods And Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 E Flair Drive, El Monte, CA 91731.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The survey system, hereinafter referred to as "instrument" or "SIAGL" shall consist essentially of a wire suspended automatic azimuth north seeking gyroscope, integrated electronics and control panel, battery power pack, top mounted scale reading theodolite and a telescopic leg tripod with provisions for mounting the assembled instrument.

3.1.1 Prior-to-assembly. Prior to assembly of the instrument, the contractor shall test and record test results of the internal components specified in 4.3.1 previous to submission of the first article instrument for inspection (see 3.2). Each subsequent instrument produced shall receive the same test, the resulting test data shall be recorded and the data sheet packed for delivery with each instrument for which the data sheet applies.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The instrument shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the instrument may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the instrument may be newly fabricated from recovered materials to the maximum extent practicable, provided the instrument produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the surveying instrument.

3.3.5 Threaded parts. Threaded parts shall be in accordance with FED-STD-H28.

3.4 Design.

3.4.1 Portability. The total weight of the instrument including battery power source, battery charger and tripod with each major component and applicable accessories in their respective transport cases shall not exceed 140 pounds. The

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weight of any individual transport case shall not exceed 42 pounds when loaded. Each component transport case shall be compatible with the Army packboard NSN 8465-00-262-5281 (see 3.21).

3.4.2 Controls. All operating controls, including data input keys and control displays shall be positioned on the instrument control panel (see 3.6.1.3). Controls shall conform to the requirements of 3.5.9.

3.4.3 Power source. The instrument shall require not more than 35 watts peak power at nominal 24 volts dc when supplied from the following sources:

a. Battery power pack, 24 volt dc.
b. Vehicle power through dc power adapter.
c. 115/230 volt ac 50/60 Hz utility power source for battery charging.
d. Vehicle power through cable directly from vehicle battery.

3.5 Environmental conditions.

3.5.1 Operating temperature range. The instrument shall determine azimuths to the accuracy specified in 3.12 in an ambient temperature range of +125 +5, -0 to -40 +0, -5 °F.

3.5.2 Storage and transit temperature conditions. The instrument shall not be damaged by storage or transit at temperatures as high as 155 °F for 4 hours and as low as -40 °F for 3 days without solar radiation.

3.5.3 Wind. The instrument shall determine azimuths to the accuracy specified herein, unsheltered, in the presence of winds to 20 ±2 miles per hour (mph), with wind gusts to 25 ±2 mph.

3.5.4 Rainfall. When in the operating position, the instrument shall not be damaged mechanically or electrically and shall show no indication of moisture penetration after exposure to rainfall of 2 ±1 inches per hour, from any direction, and any angle from vertical to 45 degrees, for a period of 5 minutes ±30 seconds. Each closed instrument case and transport case shall show no indication of moisture penetration after exposure to rainfall of 3 inches per hour for a period of 2 hours.

3.5.5 Sand and dust. When in the operating position, the instrument shall not be damaged mechanically or electrically and shall operate at the accuracy specified (see 3.12) when subjected to sand and dust blown by a 1750 foot per minute wind for 30 minutes ±30 seconds. The transit case for each major component of the system, with applicable accessories, shall show no indication of sand penetration when exposed to sand and dust blown by a 1750 foot per minute wind for 6 hours ±5 minutes.

3.5.6 Shock and vibration. The instrument and accessories shall be designed and assembled so that no fixed part shall become loose, no movable part or permanently set adjustment shall shift its setting or position and no degradation shall be caused in the performance specified herein when each major component and related accessories are stored in their respective transport case and subjected to shock tests specified in 4.7.6.1 and vibration tests specified in 4.7.6.2.

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3.5.7 Magnetic field. The instrument shall perform as specified herein in the presence of a magnetic field of 3 ±1/2 gauss.

3.5.8 Safety. All parts that are energized electrically shall be fully enclosed or guarded. Warning plate or instructions shall be provided where a hazard may exist. Any protective devices shall not impair the operating function of the instrument.

3.5.8.1 Electrical equipment. Electrical equipment shall meet the criteria set forth in MIL-STD-454.

3.5.9 Human factors engineering. The instrument shall conform to accepted human factors engineering design criteria as specified in MIL-STD-1472. Emphasis shall be given, but not limited to general requirements (4), control/display integration (5.1), visual displays (5.2), controls (5.4), labeling (5.5), anthropometry (5.6), design for maintainer (5.9), small systems and equipment (5.11), and hazards and safety (5.13), as applicable.

3.6 Construction.

3.6.1 Azimuth gyroscope. The wire suspended azimuth gyroscope shall be capable of automatically determining azimuths relative to geographic north or grid north for display readout or remote integration through an RS 232 interface provision on the control panel (see 3.6.1.3). The gyroscope unit shall consist of a gyroscope motor, electronic operator control and display panel, battery pack or power unit and internal electrical power circuitry for both the gyroscope and theodolite illumination. The gyroscope shall be a box type unit with provisions for mounting and accepting a top mounted theodolite and a battery pack or power unit. The gyroscope shall be adaptable to a telescopic leg tripod specified in 3.7. A means for plumbing shall be provided on the base of the gyroscope housing. The gyro assembly shall self-cage if overturned.

3.6.1.1 Gyro motor. The gyro motor shall have an operating speed capable of producing inertial forces necessary to determine precession resulting from the earth's rotation. The gyro shall also be provided with a clamping or self-locking system to prevent equipment damage and angular displacement.

3.6.1.2 Electronic control unit. The electronic control unit shall contain within a housing the necessary mechanical, electrical and electronic equipment for the functional operation of the azimuth gyroscope. The unit shall be capable of allowing for correction constants due to azimuth deviations as a result of initial calibration of the gyroscope over a first order reference point. Constants shall be recorded on data sheet as supplied with each gyroscope (see 3.1.1). At nominal 24 volt dc power input, not more than 35 watts of electrical power shall be required for normal instrument operation.

3.6.1.3 Electronic operator control and display panel. An operator control panel shall be provided and located on the outside portion of the unit housing. The panel shall contain all function keys (see 3.4.2), an alpha numerical display unit, receptacle to accommodate both RS 232 and interface with the Global Positioning System (GPS) and use of diagnostic test equipment. The RS 232 receptacle shall be labeled "RS 232". Warning and indicator lights as required for the normal operation of the instrument shall also be located on the panel.

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3.6.1.4 Battery power pack. A 24 volt dc nickel-cadmium battery pack shall be furnished. The battery pack shall have sufficient ampere-hour capacity for the instrument to accurately conduct not less than 40 azimuth determinations at 70 °F ambient temperature. The pack shall be mounted on and adapted to the azimuth gyroscope in such a manner that will eliminate the use of external wiring. The battery pack shall be self-contained and shall be easily removable when being stored or exchanged for a newly charged battery power pack. The battery pack shall be designed such that the pack can not be installed with the polarity reversed.

3.6.1.5 Circuit monitoring points. The gyroscope shall be provided with a built-in system test arrangement for obtaining fault analysis prior to making azimuth determinations. The built-in test system shall be capable of detecting and locating failures down to component level.

3.6.1.6 Panel illumination. The operating panel shall be equipped with an adjustable lighting system. Under nighttime operating conditions, there shall be no visible light beyond 30 meters. The lights shall be adjustable from 0.07 to 0.35 cd/m^2. The light adjustment control shall conform to 3.5.9 except low luminance green light shall be provided.

3.6.1.7 Panel markings. Each operating key, indicator light, switch or receptacle on the control panel shall be clearly marked or identified by its functional description in accordance with 3.5.9. System names, model numbers or registered trade names shall be restricted to the instrument identification plates.

3.7 Tripod. A collapsible type telescopic leg tripod shall be furnished. The tripod shall conform to MIL-T-14091 except the tripod shall be provided with a 3-point standard screw type adjustable plate assembly with a circular center opening of sufficient size to permit a 1-inch diametrical adjustment for the instrument over an established ground point. The plate assembly shall be provided with a circular level for rough leveling and instrument set-up. The adapter plate, when mounted on the tripod, shall be included in the overall height of the tripod. The tripod shall have an adjustable height range from 35 ±2 to 53 ±2 inches. The tripod shall be as lightweight as possible, consistent with the stability and overall weight requirements of the instrument.

3.8 Theodolite.

3.8.1 Description. The theodolite shall be an optical reading directional instrument, with a least circle graduation of 0.2 mm, lightweight (not more than 13 pounds), rugged and compact. The theodolite shall consist essentially of an alidade, a microscopic optical system for reading the horizontal and vertical glass circles, tangent screws, levels and a built-in auto-collimation for azimuth transfer from the gyroscope. Accessory equipment compatible with the system shall be furnished and shall include an eyepiece prism, an eyepiece sun filter, a lens brush, plumb bob assembly, and a circle adjusting wrench.

3.8.2 Telescope. The telescope shall be erect image, internal-focusing, dustproof but not pressure tight. The internal focusing tube shall move within the main tube without binding. Backlash shall not be evident when focusing the telescope. The focus shall not change when the telescope is transited. The

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telescope shall transit at the objective end with the prismatic eyepiece attached. The telescope shall be balanced when focused at infinity.

3.8.3 Sights. The telescope shall be provided with external sights aligned so that, when the telescope is pointed in the direct position at any target, the target shall appear in the field of view of the telescope.

3.8.4 Optical system. The optical system shall produce a sharp image throughout the entire focusing range and shall be free from internal reflections, spherical and chromatic aberration, coma, astigmatism and distortion. The field of view shall be substantially flat from the center to a zone outside the stadia lines equal to one-fourth of the stadia interval and shall show no marked curvature from there to the edge of the field of view. The principal axis of the lens system shall be in alignment and coincide with the collimation of the telescope in adjustment. Other optical properties shall be as specified in table I.

TABLE I. Optical properties.

Magnification	30 X
Effective aperture of objective	42 mm nominal
Field of view at 1000 ft/m.	27 ft/m
Focus, min.	1.7 ±0.1 meter
Limiting angle of resolution	
Maximum	4.2 Seconds

3.8.4.1 Telescope eyepiece. The telescope shall be fitted with an eyepiece constructed so that the eye of an observer can be placed in a position coincident with the exit pupil. The eyepiece shall be provided with a diopter scale plainly marked in one diopter increments from -4 to +4. When focusing on the reticle, the eyepiece shall operate without binding or backlash but with a firmness that will retain the focus.

3.8.4.2 Telescope eyepiece filter. When specified (see 6.2), a telescopic eyepiece filter shall be furnished for solar observations and shall be attachable without the use of tools. The telescope eyepiece filter shall contain an absorption-type black or Nichrome-coated or equal neutral filter with a density range of between 5.0 and 6.0. When a filter is furnished, the coated surface shall be toward the eyepiece lens.

3.8.4.3 Telescope eyepiece prism. When specified (see 6.2), a telescopic eyepiece prism shall be furnished to permit positioning for viewing without disturbing the focus of the telescope eyepiece. The prism shall permit observation of ground points and angles within 26 degrees from the zenith with the telescope both in the direct and reversed position. The reticle pattern shall be clearly visible and the magnification change shall not exceed 2 percent for any position of the prism.

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3.8.4.4 Eyepiece prism filter. When specified (see 6.2), an eyepiece prism filter shall be furnished. The eyepiece prism filter shall contain an absorption-type black or Nichrome-coated or equal neutral filter having a density range between 5.0 and 6.0. When a filter is furnished, the coated surface shall be placed adjacent to the telescope eyepiece.

3.8.4.5 Auto-collimation eyepiece. When specified (see 6.2), an auto-collimation eyepiece shall be furnished. The eyepiece shall be interchangeable with the normal eyepiece without the use of tools. The magnification of the telescope with this eyepiece shall be not less than 18 diameters.

3.8.4.6 Reticle. The reticle shall consist of a pattern of opaque lines registered on optical glass. The crosslines shall extend to the edge of the field of view. The single vertical crossline shall subtend equal angles with the adjacent double lines within ±0.5 seconds. The width of the lines shall be 2.5 to 4.0 microns. The double vertical lines shall subtend an apparent angle of 40 +5, -10 seconds. The radius of the solar circle shall be 15 minutes 30 ±10 seconds. The horizontal and vertical stadia lines shall be spaced so that 1/100 of the distance from a stadia rod at a known distance greater than 100 feet (30 meters) from the instrument shall be subtended with an error not to exceed 0.3 percent. The lines shall subtend 5 to 7 minutes of arc in length and shall be parallel to the corresponding crosslines. The reticle shall be mounted in a metal cell which shall not become distorted during adjustment of the collimation. The horizontal and vertical collimation adjustment of the reticle shall be positive and easily accessible. The horizontal and vertical collimation errors shall be not more than 0.2 minute (0.06 mm).

3.8.4.7 Coated optics. All internal lens-air surfaces of the telescope except the registered side of the reticle shall be coated with a reflection-reducing film in accordance with MIL-C-675 as a minimum.

3.8.5 Circles. Horizontal and vertical circles shall be made of glass. The graduations and numbers shall be opaque, legible, permanent, and silhouetted against a light background. The circles shall be fastened to rigid supports without introducing strains which will distort, chip, or break the glass. The circles shall read directly to 0.2 mm.

3.8.5.1 Horizontal circle. The horizontal circle shall be graduated every 0.2 mm and numbered every 10 mm. The numbering shall be from 000 to 639. The horizontal circle shall be mounted so that readings increase as the alidade is rotated clockwise. The root mean square (RMS) error of the circle shall not exceed 0.2 mm at ambient temperature of approximately +70 °F.

3.8.5.2 Breakdown scale. The breakdown scale shall be fixed, 10 mm long, and shall be graduated at 0.2 mm intervals. The graduations shall be numbered at 1 mm intervals. The numbering shall begin with 0 mm and shall be consecutive through 10 mm.

3.8.5.3 Vertical circle. The vertical circle shall be graduated and numbered in the same manner as the horizontal circle with zero representing zenith and reading of 1,600 mm obtained when the telescope is level and in the direct position.

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3.8.6 Microscope. A microscope shall be provided through which the horizontal and vertical circles shall be viewed. The reading section of both the horizontal and vertical circles shall be visible simultaneously through the reading microscope. The images of both horizontal and vertical circles and index lines shall appear erect and in the same focal plane when viewed through the microscope eyepiece. The image of the vertical circle shall be above the corresponding image of the horizontal circle. A different color between the horizontal circle and vertical circle is permissible.

3.8.6.1 Illumination. The theodolite shall be illuminated by both natural lighting and electrical means. Natural illumination shall be provided by use of mirrors. Electrical illumination shall be provided by use of lighting attachments. The lighting attachments shall be equipped with a frosted glass cover between the light source and the theodolite opening. When the theodolite is operated in the dark, there shall be no light visible beyond 30 meters except that within the telescope field of view.

3.8.6.1.1 Natural lighting. Adjustable mirrors, with a reflective material coating on the second surface, shall be provided to illuminate the vertical and horizontal circles and corresponding breakdown scales. Moving or adjusting the mirrors shall not cause a variation in the circle reading of more than 0.03 mm of arc.

3.8.6.1.2 Electrical lighting. Lighting attachments, when used, shall illuminate the horizontal and vertical circles along with the corresponding breakdown scales. A means for varying the intensity of the light shall be provided. Illumination shall conform to 3.5.9.

3.8.6.2 Power. Power for illumination of the theodolite scales and reticle shall be provided through a cable attached to the auto-collimation unit. Illumination of the reticle shall introduce minimum adverse effect on the optical properties of the telescope. Intensity adjustment shall be provided.

3.8.6.3 Wiring. The theodolite shall be wired as required for the operation of the lighting equipment. Wiring shall allow sufficient slack between terminals for repair of broken connections. Slack wire shall not interfere with the operation of the theodolite.

3.8.6.4 Electric lamps. The electric bulbs shall be a screw base with frosted glass. Bulb shall be rated at 2.5 V dc except light emitting diodes (LED).

3.8.7 Prisms and mirrors. Any metal surfacing on prisms or mirrors shall be tarnishproof. Front-surface mirrors may be substituted for prisms provided sufficient illumination is available for reading of the scales. Mirrors shall be either aluminized optical flats protected by an aluminum oxide film or a front-surface, tarnishproof chrome plate.

3.8.7.1 Mountings. All prisms, mirrors and lenses used in the optical system by which the circles are read shall be mounted so they will not shift. Optical components shall be free from strains within the mountings.

3.8.7.2 Standards. The standards shall be factory adjusted and permanently set.

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3.8.7.3 Bearings. The telescope shall be supported on bearings to provide precise orientation of the telescope. The bearings shall provide true centering with a minimum of strain or drag.

3.8.8 Levels. The theodolite shall be equipped with a graduated plate level. Level vial graduations shall be clear, uniform, wear-resistant and symmetrical about the exposed portion and shall extend far enough to include the ends of the bubble at any ambient temperature from +125 to -40 °F without solar radiation. Movement of the level bubbles shall be uniform and free of abrupt changes.

3.8.8.1 Plate level. The plate level shall be graduated to 2 mm spacing and shall have a sensitivity of 30 ±8 arc seconds for each spacing. The level shall be mounted directly over the vertical axis to provide maximum stability, protection and rapid leveling. A clear plastic cover shall be provided to reduce sensitivity of the level vial to radiated heat.

3.8.8.2 Altitude self-indexing device. The altitude self-indexing device shall be damped so that oscillation is stopped in not more than 5 seconds. The device shall be accurate within 0.06 mm.

3.8.8.3 Operating screws. Where applicable, leveling screws and tangent screws shall have a take-up for wear adjustment. The horizontal and vertical motion tangent screws shall be long enough to produce 100 mm of motion over their entire run. Tangent screws shall be removable for cleaning.

3.8.9 Instrument cover. A protective cover for the instrument shall be provided. The cover shall be rainproof and constructed of a lightweight material capable of withstanding the operating temperature range (see 3.5) without damage. The cover shall fit over the instrument when the instrument is set up for operation.

3.8.10 Target rod. When specified (see 6.2), a target rod shall be furnished. The rod shall be adaptable to the top of the theodolite. The target rod shall be cylindrical in shape with a provision to light the top of the rod for nighttime observations. The color of the illumination shall be capable of changing. The top of the rod shall be within ±1.5 mm of the projected vertical axis of the theodolite. Power for the illumination of the target rod shall be furnished by a short cable connected to the auto-collimation box receptacle.

3.9 Cables and connectors. External power cables shall be provided to transmit power from both ac and dc power sources for operating the instrument and charging the instrument batteries. One each of the following power cables shall be furnished (see 3.4.3):

a. Cable from a 24-volt, dc power source (vehicle battery or a dc receptacle on vehicle) for charging instrument batteries, not less than 15 feet long.
b. Cable from a 115/220-volt, 50/60 Hz power source to battery charger, not less than 6 feet long.
c. Cable from a 24-volt, dc power source (vehicle battery or dc receptacle on vehicle) to a power conditioner or adapter when used in lieu of regular battery power pack, not less than 25 feet long.

Each cable furnished shall be equipped with connectors and adapters as required for their intended application.

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3.10 Maintainability. When the instrument is tested as specified in 4.7.9, the ratio of maintenance man-hours to operating hours shall not exceed 0.05. Each operation as specified in 4.7.9 shall be accomplished by one man using the tools furnished with the instrument.

3.11 Transportability.

3.11.1 Cases. Each major component of the instrument assembly shall be provided with a protective case for storage and transport. Each case shall be constructed of materials capable of protecting each component and its related accessories from shock, vibration and impact (see 3.5.6) that may be encountered during transport. Each case shall be moisture and dustproof, and equipped with a carrying handle. Adjustable carrying straps with attached cushion material shall be provided to permit backpacking each case individually or with cases mounted on the Army packboard (see 3.4.1 and 3.21). The carrying straps shall be adjustable to fit the 5th percentile female and the 95th percentile male soldier (see 3.5.9). Each case shall contain its assigned component and related equipment, tools, accessories and instruction manuals as required for assembly and operation of the instrument. Each case shall be permanently marked to indicate that it contains a delicate instrument and show the proper orientation of the case for transport. All cases shall provide adequate protection for each component during shipment and all storage conditions.

3.12 Performance. The instrument shall determine azimuths at any operational latitude up to 70 degrees with an error in true azimuth determination not to exceed 0.2 mm probability of error (PE) and an error not to exceed 0.3 mm PE from 70 to 75 degree latitudes. The time required from power-on to determination of azimuths of required accuracy shall not exceed 3.5 minutes for all latitudes up to 75 degrees at ambient temperatures specified in 3.5. Preorientation or instrument warm-up shall not be required. The instrument shall permit removal from the operating site within 2 minutes without affecting the accuracy of the gyroscope sensor. The instrument shall be capable of not less than 100 missions of operation without recalibration, alignment, or adjustment to a predetermined azimuth. The instrument shall be provided with a level vial of the required accuracy and sensitivity to permit the instrument to be set up and operated automatically without interruption. The instrument shall self-cage if overturned or subjected to a sudden shock or vibration.

3.13 Electromagnetic interference. The electromagnetic interference emission characteristics of the instrument shall conform to MIL-STD-461, with tests limited to those specified in 4.7.12.

3.14 Fungus- and moisture-resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.14.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus- and moisture-resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

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When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.14.1 Composition. Composition II shall be used only in the case when local air pollution regulations in the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with the South Coast Air Quality Management District, Rule 442, Usage of Solvents.

3.15 Treatment and painting. Unless otherwise specified (see 6.2), the portions of the instrument normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, color olive drab No. 24064 in accordance with FED-STD-595 except the transport case, transit case, electronic control unit case and instrument tripod coating shall be in accordance with MIL-C-22751 using the same color.

3.16 Identification marking. The instrument shall be identified in accordance with MIL-STD-130. The marking shall be applied to the instrument on plates conforming to MIL-P-19834, type II. Plates shall be attached in a conspicuous protected location.

3.17 Winterization kit. When specified (see 6.2), a winterization kit shall be furnished for the instrument. The winterization kit shall assure operation of the instrument at temperatures specified in 3.5.1. The winterization kit shall consist of the following items:

	Item	Quantity
a.	Lens brush	1
b.	Chamois leather	1
c.	Lens tissue	3 pkgs
d.	Orangewood sticks	Pkg of 25

3.18 Tool and accessory case. The case shall be made of cotton conforming to CCC-C-419, type 1 No. 12, olive drab color, rendered fungus resistant and water repellent and shall be equipped with a slide fastener in accordance with V-F-106. The case shall be sized and equipped with pockets to contain and provide proper storage for the following items:

a. Winterization kit (see 3.17).
b. Instrument cover (see 3.8.9).
c. Container, lubricant.
d. Prism, eyepiece, telescope.
e. Prism, eyepiece, microscope.
f. Filter, solar.
g. Filter, prism.
h. Pins, adjusting.
i. Screwdriver, flat tip.
j. Screwdriver, jewelers.

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3.19 Lubrication. Prior to assembly, the theodolite shall be lubricated with thermally stable lubricant.

3.20 Workmanship. All threads requiring movement, and sliding and rotatable parts, shall move without binding, backlash, or end play. Clamping actions shall be positive and, when unclamped, shall not cause drag. Focusing components and assemblies shall retain their focus when finger pressure is released, and the telescope shall not change focus when transited. No lateral motion shall exist between the telescope and standards. Lenses shall be free of scratches, chips, air bubbles between cemented elements, dirt between the glass and coated film, or cloudiness. Optics shall show sharpness of focus and distinct images to provide favorable definition. Metal surfacings on the optical components for all prisms and lenses shall not shift. There shall be no flickering of the illumination system when the vertical or horizontal motions are rotated. Liquid within the level vials shall be clear, and the vials shall be tight in the housings. Components and assemblies shall show no evidence of forced attachment. Bearing surfaces shall show no scratches or surface blemishes. Mirrors shall not be tarnished, and backing shall be uniform and unspotted.

3.21 Government loaned property. When specified (see 6.2), the following property in the quantity indicated will be loaned by the Government (see 6.7).

Item No.	Description	Quantity
1	Army packboard NSN 8465-00-262-5281	As required

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

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4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Prior-to-assembly inspections (see 4.3).
b. First article inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Inspection of packaging (see 4.9).

4.3 Prior-to-assembly inspection.

4.3.1 Tests. Prior to assembly, the horizontal and vertical circles shall be tested for accuracy, the level vials shall be tested for sensitivity and the reticles shall be measured for reticle line weight as specified in 4.3.1.3. Failure of the applicable test shall be cause for rejection of the applicable items.

4.3.1.1 Circles. Horizontal and vertical circles shall be tested for accuracy. In carrying out these tests, the circles shall be mounted on a precise turntable with fixed microscopes located so that the comparison can be made of nominally equal angles on different parts of the circle. Any other method of test giving equal results is acceptable. Evidence of a graduation error exceeding 3 seconds or 0.015 mm for horizontal circles or 3 seconds or 0.015 mm for vertical circles shall constitute failure of this test.

4.3.1.2 Level sensitivity. The sensitivity of each level vial shall be determined by mounting the level vial on a level trier or by any other method yielding values accurate to within 5 percent. The plane of the graduated vials shall be changed a sufficient amount to cause a change of not less than two graduations in the position of the level bubble. The plane of the ungraduated vials shall be changed a sufficient amount to cause a change of not less than 4 mm in the position of the level bubble as indicated on a scale placed adjacent to the vials. Evidence that the levels do not have the sensitivities specified herein shall constitute failure of this test.

4.3.1.3 Reticle line weight. The reticle line weights shall be determined by using a measuring microscope or by using any other method that will yield comparable values. Evidence of the reticle line weights differing from 2.5 to 4.0 microns shall constitute failure of this test.

4.3.1.4 Theodolite system test (horizontal) with two collimators. This test is based on a series of observations on two collimators approximately 36 degrees (640 mm) apart and on a plane having an elevation approximately the same as that of the horizontal axis of the instrument being tested. In making the following observations, each circle reading shall be estimated to the nearest tenth of a graduation of the breakdown scale. This test measures the following factors pertaining to the theodolite:

a. Ability of the instrument to repeat measurements.
b. Average circle graduation error.
c. Collimation error.
d. Circle eccentricity (horizontal circle).

Two sets of measurements shall be made. Each set shall consist of 20 angles (40 measurements) by using all sections of the horizontal circle. A copy of the

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completed test data sheets or computer printout of same shall be inclosed with the first article instrument. Absence of data sheets shall constitute failure of this test.

4.4 First article inspection.

4.4.1 Examination. The first article instrument shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. The first article instrument shall be subjected to the tests marked "X" in column 1 of table II. Failure of any test shall be cause for rejection.

4.5 Quality conformance inspection.

4.5.1 Sampling. Samples for quality conformance shall be selected in accordance with MIL-STD-105.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.3 Tests. Samples selected in accordance with 4.5.1 shall be subjected to tests marked "X" in column 2 of table II. One or more failures shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination.

Major

101. Materials not as specified (see 3.3).
102. Material deterioration prevention and control not as specified (see 3.3.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
104. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.3.3).
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the instrument (see 3.3.4).
106. Threaded parts not as specified (see 3.3.5).
107. Instrument total weight with cases not as specified (see 3.4.1).
108. Control panel keys, displays and markings not as specified (see 3.4.2 and 3.6.1.7).
109. Instrument operating electrical power sources not as specified (see 3.4.3 and 3.8.6.2).
110. Safety and protective devices not as specified (see 3.5.8).
111. Human factors engineering not as specified (see 3.5.9)
112. Instrument construction and assembly not as specified (see 3.6 and 3.8.7.1 thru 3.8.7.3).
113. Data sheets not provided as specified (see 3.6.1.2).
114. Theodolite mounting not as specified (see 3.6.1).

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115. Wiring not as specified (see 3.6.1, 3.8.6.3 and 3.9).
116. External power connectors not as specified (see 3.6.1.4).
117. Circuit monitoring points not as specified (see 3.6.1.5).
118. Control panel illumination not as specified (see 3.6.1.6).
119. Tripod not as specified (see 3.7).
120. Theodolite components not as specified (see 3.8.1).
121. Telescope does not balance or transit as specified (see 3.8.2).
122. Telescope optical system does not produce a sharp image through the focusing range or is not free from internal reflections, aberration, coma, or distortions (see 3.8.4).
123. Eyepiece does not retain focus (see 3.8.4.1).
124. Telescope eyepiece filter, prism, and prism filter not as specified (see 3.8.4.2, 3.8.4.3, and 3.8.4.4).
125. Auto-collimation eyepiece not as specified (see 3.8.4.5).
126. Reticle lines do not appear opaque or of uniform width (see 3.8.4.6).
127. Internal lens surfaces not coated with reflection-reducing film (see 3.8.4.7).
128. Circle accuracy not as specified (see 3.8.5.1).
129. Horizontal and vertical circles not as specified (see 3.8.5.1 and 3.8.5.2).
130. Breakdown scales not as specified (see 3.8.5.2).
131. Microscope not as specified (see 3.8.6)
132. Illumination not as specified (see 3.8.6.1 thru 3.8.6.1.2).
133. Electric lamps not as specified (see 3.8.6.4).
134. Prisms and mirrors not as specified (see 3.8.7).
135. Levels not as specified (see 3.8.8 and 3.8.8.1).
136. Level vial graduations not clear or uniform (see 3.8.8).
137. Altitude self-indexing device not as specified (see 3.8.8.2).
138. Operating screws not as specified (see 3.8.8.3).
139. Instrument protective cover not as specified (see 3.8.9).
140. Target rod not as specified (see 3.8.10).
141. Maintainability not as specified (see 3.10).
142. Transport and storage cases not as specified (see 3.11.1).
143. Fungus- and moisture-resistance not as specified (see 3.14 and 3.14.1).
144. Treatment and painting not as specified (see 3.15).
145. Identification and marking not as specified (see 3.16).
146. Lubrication not as specified (see 3.19).
147. Workmanship not as specified (see 3.20).

Minor

201. Carrying slings not as specified (see 3.11.1).
202. Winterization kit not as specified (see 3.17).
203. Maintenance tools and tool kit not as specified (see 3.18).

4.6.2 Tests. The instrument accessories, associated equipment, and transport and storage cases shall be tested as specified herein and in accordance with schedule specified in table II. Prior to testing, the contractor shall adjust and calibrate the instrument by established methods as required to operate as specified herein. After adjustment of the instrument by the contractor, readjustment, if necessary, may be made only at the conclusion of the shock, vibration, and extreme environmental tests.

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TABLE II. Inspection schedule.

First Article	Quality Conformance	Test	Requirement Paragraph	Test Paragraph
X		Prior-to-assembly	3.1.1	4.3
X		Circles	3.1.1	4.3.1.1
X		Level sensitivity	3.1.1	4.3.1.2
X		Reticle line weight	3.1.1	4.3.1.3
X		Theodolite system	3.1.1	4.3.1.4
X		Portability	3.4.1	4.6.3
X		Power sources	3.4.3	4.6.4
X		High latitude accuracy	3.12	4.7.1
X		Instrument operation	3.5	4.7.2
X		Operating temperature range	3.5.1	4.7.2.1
X		Wind	3.5.3	4.7.3
X		Rainfall	3.5.4	4.7.4
X		Sand and dust	3.5.5	4.7.5
X		Shock and vibration	3.5.6	4.7.6
X		Magnetic field	3.5.7	4.7.7
X		Battery power pack	3.6.1.4	4.7.8
X		Maintainability	3.10	4.7.9
X		Transportability	3.11	4.7.10
X	X	Performance	3.12	4.7.11
X		Electromagnetic interference	3.13	4.7.12

4.6.3 Portability. With each major component and related accessories placed in the respective transport case, (see 3.11.1) determine the weight of each case and tripod. Using transport case shoulder straps (see 3.11.1), secure each case or cases and tripod on one or more packboards as required. Nonconformance to 3.4.1 or 3.11.1 shall constitute failure of this test.

4.6.4 Power and power sources. Assemble and set-up the instrument as required to make an azimuth determination. With both the power and panel illumination switches activated, along with external lighting attachments for night-time operation, simulate an azimuth determination. Using an appropriate electrical instrument, determine the maximum power consumption rate recorded during this test. Power required in excess of 35 watts or nonconformance to 3.4.3 or 3.6.1.2 shall constitute failure of this test.

4.7 Environmental conditions.

4.7.1 High latitude accuracy. With the first article instrument previously adjusted and calibrated (see 4.5.2), conduct this test at the Government test facility, Fort Greely, Alaska (see 6.4) at approximately 64-70 degrees north latitude with the gyroscope reference unit sheltered and under the existing ambient temperature at the time of test. No adjustments, calibration, or maintenance shall be performed at the test site or upon return except what is normally performed as part of operating procedure or operator maintenance. Position the instrument over a point for which a first order astronomic azimuth has been established. Perform not less than 10 missions as specified in 4.6.11. Failure of test specified in 4.7.11 or nonconformance to 3.12 shall constitute failure of this test.

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4.7.2 Instrument operation. An operational test shall be conducted following each of the environmental, shock, and vibration tests as specified herein. The instrument shall be operated and checked to determine resulting operational deficiencies as follows:

a. Manipulate all movable parts and check for freedom of movement and noninterference with other parts in various operating positions throughout their maximum ranges.
b. Fit or attach all accessories to the instrument and check for functionality as specified.
c. Check the levels for free movement to determine if the bubble remains within the graduations.
d. Rotate the theodolite about its horizontal and vertical axes and check the illumination system for uniform and steady operation.
e. Check microscope focus for uniform and clear focus around each entire circle.
f. Check parts, components, and accessories for damage.

Evidence that the instrument, components, or accessories will not function as specified, presence of broken, cracked, or shifted parts of the instrument or accessories, presence of splinters or cracks, broken, cracked, or loosened hardware, opened joints, shifted padding, or deformation of the cases shall constitute failure of this test.

4.7.2.1 Temperature range.

4.7.2.2 High-temperature operation. A previously calibrated SIAGL shall be subjected to a high-temperature test in accordance with MIL-STD-810, method 501, procedure II, with an operating temperature of +125 +5, -0 °F. After stabilizing at not less than +125 °F, the instrument shall be removed from its transport case and tested in accordance with 4.7.2. Evidence of damage to operating parts or accessories, or nonconformance to 3.5.1 or 3.5.2 shall constitute failure of this test.

4.7.2.3 High-temperature storage. A previously calibrated SIAGL shall be subjected to a high-temperature test in accordance with MIL-STD-810, method 501, procedure II, with a storage temperature of +155 +5, -0 °F. The instrument shall be stored in its transport case. At the end of the test, the instrument shall be removed from its transport case and tested in accordance with 4.7.2. Evidence of damage to operating parts or accessories, or nonconformance to 3.5.1 or 3.5.2 shall constitute failure of this test.

4.7.2.4 Low-temperature operation. A previously calibrated SIAGL shall be subjected to a low-temperature test in accordance with MIL-STD-810, method 502, procedure I, with an operating temperature of -40 +0, -5 °F. After the completion of the operational test, the instrument will be subjected to the test specified in 4.7.2. Evidence of damage to operating parts or accessories, or nonconformance with 3.5.1 or 3.5.2 shall constitute failure of this test.

4.7.2.5 Low-temperature storage. A previously calibrated SIAGL shall be subjected to a low-temperature storage test in accordance with MIL-STD-810, method 502, procedure I, with an storage temperature of -40 +0, -5 °F. The instrument will be stored in its transport case during a 72-hour storage period. After completion of the storage period the temperature shall be raised, the instrument

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shall be removed from its transport case, and tested in accordance with 4.7.2. Evidence of damage to operating parts or accessories, or nonconformance to 3.5.1 or 3.5.2 shall constitute failure of this test.

4.7.3 Wind. Test the instrument on the 35-inch tripod height in accordance with 4.7.11 while the instrument is subjected to simulated wind. The wind shall be directed horizontally toward the instrument panel, parallel to and perpendicular to the spin axis of the gyro. Wind speed shall be not less than 20 ±2 mph steady state with gusts to 25 ±2 mph at a rate of 5 to 15 gusts per minute. The accuracy and time from turn-on to completion of an azimuth measurement shall be determined. Nonconformance to 3.5.3 or 3.12 shall constitute failure of this test.

4.7.4 Rain test.

4.7.4.1 Assembled instrument. The instrument, when assembled in its normal operating condition, shall be placed in a rain testing chamber and tested in accordance with MIL-STD-810, method 506.2, procedure I. The test chamber shall be equipped with spray nozzles that will provide an even rain pattern over the test area at the rate of 2 ±1 inches per hour. Air velocity shall be provided to simulate wind driven rain conditions from 0 to 45 degrees from vertical. The instrument shall be subjected to this test for a period of 5 minutes. The instrument shall then be dried off and tested in accordance with 4.7.2. Evidence of damage to operating parts, internal malfunction or nonconformance to 3.5.4 or 4.7.2 shall constitute failure of this test.

4.7.4.2 Transport cases. Each transport case containing its respective component with accessories, parts, manuals and other related equipment as required for operation, shall be subjected to the rain test specified in 4.7.4.1. In addition, the cases shall be exposed to a wind source producing horizontal rain and wind at velocities of 40 ±5 mph to all sides of the case equally for a period of not less than 1 hour. Presence of water in the case, swelling or buckling of materials or inability of the cover to close or seal properly shall constitute failure of this test.

4.7.5 Sand and dust. The instrument, assembled in the normal operating configuration, shall be placed in a sand and dust chamber and tested in accordance with MIL-STD-810, method 510.2, procedure I-II. The sand and dust shall be passed through the test chamber at a velocity of 300 to 1750 feet per minute. The temperature in the chamber shall be maintained at 77 ±5 °F (25 ±2.8 °C). The relative humidity shall not exceed 30 percent. The instrument shall be subjected to this test for a period of 5 minutes ±30 seconds. The instrument shall be removed from the test chamber, wiped clean and disassembled for transport or storage. Each component shall be stored in its transport case and the cover closed. Repeat the above test procedure with each component stored in its respective transport case for a period of 6 hours ±5 minutes. Remove the components from their transport cases, reassemble the instrument and perform the operational test specified in 4.7.2. Evidence of sand or dust penetration in the case, instrument malfunction or nonconformance to 3.5.5 or 4.7.2 shall constitute failure of this test.

4.7.6 Shock and vibration.

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4.7.6.1 Shock. Each major component of a previously calibrated instrument, stored in its respective transport case, shall be tested in accordance with MIL-STD-810, method 516.3, procedure III, with the following exceptions:

a. Shock pulse shall be 30 g's - 11 millisecond of figure 516.3-1.
b. Height of corner drop will be 13 inches.

The instrument shall be reassembled and tested in accordance with 4.6.2. Nonconformance to 3.5.6 or 4.7.2 shall constitute failure of this test.

4.7.6.2 Vibration. Each major component of a previously calibrated instrument, stored in transport case, shall be tested in accordance with MIL-STD-810, method 514.3, procedure I, figures 514.3-1 thru 514.3-3. The instrument shall be vibrated in 3 mutually perpendicular planes. Vibration testing in each plane shall be conducted at +155, -65, and +75 °F ambient temperatures. The instrument shall then be reassembled and tested in accordance with 4.7.2. Nonconformance to 3.5.6 or 4.7.2 shall constitute failure of this test.

4.7.7 Magnetic field. Position the instrument over a point from which a first-order astronomic azimuth has been established. Position a magnetic source capable of producing a magnetic field of 3 ±1/2 gauss at the instrument. Make a group of azimuth determinations with the lines of magnetic force directed from each of the following directions: north-south, east-west and vertical. Calculate the deviation from the true azimuth of each group of azimuth determinations. Nonconformance to 3.5.7 or 3.12 shall constitute failure of this test.

4.7.8 Battery power pack. Record number of azimuth determinations completed by a fully charged battery power pack during instrument testing specified in 4.7.11. Failure to accomplish the required azimuth determinations or nonconformance to 3.6.1.4 shall be cause for rejection of the battery power pack.

4.7.9 Maintainability. The following maintenance operations shall be conducted as required, and the time recorded during performance testing (see 4.7.11), to determine conformance to 3.10. The ratio of maintenance man-hours to operating hours accumulated during performance testing shall not exceed that specified in 3.10. The recommended maintenance operations are as follows:

a. Cleaning and checking instrument per operator's manual will be done by the operator and shall not require more than 1 hour each day.
b. Checking, adjustments, and repair actions shall not require more than 4 hours each week.
c. Checking, and if necessary, calibrating reference mirror to the gyroscope on a known base line shall not be required at an interval of less than 100 missions.

Inability of the instrument to accomplish its mission within allocated maintenance time specified, an excessive man-hour maintenance/operating ratio, or nonconformance to 3.10 shall constitute failure of this test. During the performance testing, manuals and operator/maintenance tools furnished with the instrument shall be evaluated. Inability to properly maintain the instrument using the support elements furnished shall constitute failure of this test.

4.7.10 Transportability. Each transport and storage case containing its assigned component and related accessories, when properly prepared for field

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transport, shall have previously completed the high and low temperature test specified in 4.7.2.1, rain test specified in 4.7.4, and the shock and vibration test specified in 4.7.6. The carrying straps on each transport case (see 3.11.1) shall be adjusted for field transport and each case backpacked for a period of 30 minutes by an average size individual. Using the carrying straps on each case, the cases shall then be secured to the Army packboard (see 3.4.1), and field transported. Incompatibility with the Army packboard or nonconformance to 3.4.1 or 3.11.1 shall constitute failure of this test.

4.7.11 Operational performance. Conduct this test with a previously calibrated instrument (see 4.5.2) positioned over a point or number of points from which a first-order astronomic azimuth has been established. This test shall be conducted in the ambient operating temperature range specified in 3.5.1 and within a latitude range up to 70 degrees (see 3.12). Using a fully charged battery power pack, perform not less than 25 azimuth determinations. Record the results of each determination or mission, including the time required from power-on to determination of initial azimuth, on an appropriate data sheet for performance evaluation. Instrument error, excess time required to dismantle, pack, and remove instrument from operating site or nonconformance to 3.12 shall constitute failure of this test.

4.7.12 Electromagnetic interference. The instrument shall be tested to determine conformance to 3.13. Using a regular battery power pack, and simulating a normal operating environment, the following tests shall be conducted:

CE01 - 30 Hz to 50 kHz
CE03 - 20 kHz - 20 MHz
RE02 - 14 kHz - 1.0 GHz
CS01 - 30 Hz - 50 kHz
CS02 - 50 kHz - 400 MHz
CS06 - Spike
RS01 - 30 Hz - 30 kHz (discrete frequencies 60, 120, 400 Hz)
RS02 - Magnetic induction field
RS03 - 14 kHz to 10 GHz (unsheltered)

The contractor shall furnish the contracting officer a report of the tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.8 Inspection comparison. The Government may select instruments at any time during the contract production period and subject the instruments to any examination or test specified herein or as otherwise deemed appropriate by the Government to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected first-produced model and initial production instruments. In addition to any test specified as part of the quality conformance, the Government reserves the right to conduct any and all other tests contained in this specification as part of the quality conformance, and failure of such additional tests shall have the same effect as failure of those tests specified as quality conformance.

4.8.1 Inspection failure. Failure of an quality conformance instrument to meet any requirement specified herein during and as a result of the examination and

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tests specified in 4.4 shall be cause for rejection of the quality conformance instrument(s) and shall be cause for refusal by the Government to continue acceptance of production instruments until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on instruments previously accepted and produced under the contract. Any deficiencies found as a result of the quality conformance will be considered prima facie evidence that all instruments accepted prior to the completion of quality conformance are similarly deficient, unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.9 Inspection of packaging. The preservation, packing, and marking shall be examined and tested in accordance with the quality assurance provisions of MIL-S-3534.

5. PACKAGING

5.1 Preservation, packing, and marking. Each complete instrument with all components and accessories provided, shall be preserved and packed in accordance with MIL-S-3534. The level of preservation and packing shall be level A, B, or C as specified (see 6.2). Marking for levels A, B, and C shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The instrument is intended for use in establishing a reference line from which azimuths to various points may be quickly determined in artillery fire-control and missile-laying surveys.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. When first article is not required (see 3.2).
d. When telescope eyepiece filter is required (see 3.8.4.2).
e. When telescope eyepiece prism is required (see 3.8.4.3).
f. When eyepiece prism filter is required (see 3.8.4.4).
g. When auto-collimation eyepiece is required (see 3.8.4.5).
h. When target rod is required (see 3.8.10).
i. Treatment and painting required when other than specified (see 3.15).
j. When winterization kit is required (see 3.17).
k. When property will be loaned by the Government (see 3.21).
l. Level of preservation and packing required (see 5.1).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the

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first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 High latitude test. The contracting officer should make arrangements for the contractor to use Fort Greely, Alaska, facility for the first article test specified in 4.6.1.

6.5 Condition for inspection. The contracting officer should require the contractor to certify that the instrument(s) is/are ready for inspection as specified in table II.

6.6 Product improvements. The contractors latest product improvements incorporated in their current model surveying instrument along with associated accessories shall be furnished. Where specific requirements are not stated herein, all items listed in the manufacturers' published specifications, brochures, and catalogs which are normally furnished to the commercial customer as standard equipment, shall also be furnished.

6.7 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.21.

6.8 Subject term (key-word) listing.

Gyroscope, north seeking azimuth
Instrument, topographic survey
Instrument, artillery fire-control
Instrument, missile-laying survey
"SIAGL"
Theodolite

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Review activity:
DLA - GS

Preparing activity:
Army - ME

Project 6675-0903

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-52256C(ME)	90/08/24

3. DOCUMENT TITLE
Surveying Instrument: Azimuth Gyro, Lightweight

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

[redacted]

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717

c. ADDRESS (Include Zip Code)
US Army Belvoir RDE Center
ATTN: STRBE-TSE
Ft. Belvoir, VA 22060-5606

IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:
Defense Quality and Standardization Office
5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466
Telephone (703) 756-2340 AUTOVON 289-2340

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